Module 4: Multiple Linear Regression

Other consideration part 1

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Other considerations/potential issues in MLR

- Dealing with categorical or qualitative predictors.
- Multicollinearity.
- Non-constant variance of error.
- Non-linear relationships.

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Dealing with categorical or qualitative predictors

Example: credit dataset

```
> str(Credit)
'data.frame': 400 obs. of  11 variables:
 $ Income   : num  14.9 106 104.6 148.9 55.9 ...
 $ Limit    : num  3606 6645 7075 9504 4897 ...
 $ Rating   : num  283 483 514 681 357 569  ...
 $ Cards    : num  2 3 4 3 2 4 2 2 5 3 ...
 $ Age      : num  34 82 71 36 68 77 37 87 66 41 ...
 $ Education: num  11 15 11 11 16 10 12 9 13 19 ...
 $ Own      : Factor w/ 2 levels "No","Yes": 1 2 1 2 1 1
 $ Student  : Factor w/ 2 levels "No","Yes": 1 2 1 1 1 1
 $ Married  : Factor w/ 2 levels "No","Yes": 2 2 1 1 2 1
 $ Region   : Factor w/ 3 levels "East","South",..: 2 3 3
 $ Balance  : num  333 903 580 964 331 ...
```

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Dealing with categorical predictors

Suppose we want to predict Balance using predictors Income and Region. How should we write out our model?

$$\text{Balance} = \beta_0 + \beta_1 \cdot \text{Income} + \beta_2 \cdot \underbrace{\text{Region}}_{\text{categorical}} + \varepsilon$$

Dealing with categorical predictors

For each qualitative predictor: Region = west, south, east

- Choose a baseline category (R will automatically choose one for you, but you can change this). east

- For every other category, define a dummy variable (also called indicator functions).

$$\text{Balance} = \beta_0 + \beta_1 \cdot \text{Income} + \underbrace{\beta_2 \cdot X_{west}} + \beta_3 \cdot X_{south} + \varepsilon$$

since east is our baseline we don't need a dummy variable

$$X_{west} = \begin{cases} 1, & \text{if west} \\ 0, & \text{if not west} \end{cases}$$

$$X_{south} = \begin{cases} 1, & \text{if south} \\ 0, & \text{if not south} \end{cases}$$

$$\text{Balance} = \beta_0 + \beta_1 \cdot \text{Income} + \beta_2 + \varepsilon \quad // \text{west}$$

$$\text{Balance} = \beta_0 + \beta_1 \cdot \text{Income} + \beta_3 + \varepsilon \quad // \text{south}$$

$$\text{Balance} = \beta_0 + \beta_1 \cdot \text{Income} + \varepsilon \quad // \text{east}$$

Dealing with categorical or qualitative predictors

What if we have a categorical predictor like highest education level:

- 1 = high school diploma, → baseline
- 2 = college degree,
- 3= master's degree,
- 4= doctoral degree.

} dummy variables

In this case, do we want to create a collection of dummy variables to represent this categorical predictor?

k levels = k-1 dummy variables

Dealing with categorical or qualitative predictors

When you have categorical predictors in your model...

- the predicted values $\hat{Y}$ are independent on the choice of the baseline category.
- the interpretation of the regression coefficients (and associated tests) depend on the baseline category.
- standard model selection techniques break down.
    - To check whether a group of dummy variables is important, use an F-test.

Dealing with categorical or qualitative predictors

$$x_{i1} = \begin{cases} 1 & \text{if } i\text{th person is from the South} \\ 0 & \text{if } i\text{th person is not from the South,} \end{cases} \tag{3.28}$$

and the second could be

$$x_{i2} = \begin{cases} 1 & \text{if } i\text{th person is from the West} \\ 0 & \text{if } i\text{th person is not from the West.} \end{cases} \tag{3.29}$$

Then both of these variables can be used in the regression equation, in order to obtain the model

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \epsilon_i = \begin{cases} \beta_0 + \beta_1 + \epsilon_i & \text{if } i\text{th person is from the South} \quad X_1 = 1, X_2 = 0 \\ \beta_0 + \beta_2 + \epsilon_i & \text{if } i\text{th person is from the West} \quad X_1 = 0, X_2 = 1 \\ \beta_0 + \epsilon_i & \text{if } i\text{th person is from the East.} \quad X_1 = 0, X_2 = 0 \end{cases} \tag{3.30}$$

How to interprete β_2 ?

↳ It tells us how much higher or lower the average income is for someone from the west compared to someone from the east

south = baseline

$$x_{i1} = \begin{cases} 1 & \text{from the east} \\ 0 & \text{not from the east} \end{cases}$$

$$y_{i1} = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon = \begin{cases} \beta_0 + \beta_1 + \varepsilon_i & \text{from east } [x_1 = 1, x_2 = 0] \\ \beta_0 + \beta_2 + \varepsilon_i & \text{from west } [x_1 = 0, x_2 = 0] \\ \beta_0 + \beta_2 + \varepsilon_i & \text{from south } [x_1 = 0, x_2 = \end{cases}$$

Dealing with categorical or qualitative predictors

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Dealing with categorical or qualitative predictors

	Coefficient	Std. error	t-statistic	p-value
Intercept	531.00	46.32	11.464	< 0.0001
region[South]	−12.50	56.68	−0.221	0.8260
region[West]	−18.69	65.02	−0.287	0.7740

p-values for the two dummy variables are very large, suggesting no statistical evidence of a real difference in average credit card balance between South and East or between West and East.

Should we abandon the predictor region ?

- Rather than rely on the individual coefficients, we can use an **F-test** to test $\mathrm{HO}: \beta 1 = \beta 2 = 0$.
- This F-test has a $\boldsymbol{p}$**-value** $= \mathbf{0.96}$ −> very high.
- We cannot reject the null hypothesis that there is no statistically significant relationship between balance and region.

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Consequences of multicollinearity

Refers to the situation when two or more predictors are highly correlated.

$$Y \sim X_1 + X_2 + X_3$$

- When two or more predictors are highly correlated, it makes it difficult to separate out individual effects of predictors on the response.
- Incorporating redundant information in your model.
- Given X_1 is in the model, X_2 is not helping to explain much of Y (and vice versa).

Consequences of multicollinearity

perfect correlation: $X_1, X_2 = X_1$ $Y = \beta_0 + \beta_1 \cdot X_1 + \beta_2 \cdot X_2 + \varepsilon$

Least Square (find $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2$)

y	X_1	X_2
2	1	1
3	1.5	1.5
6	3	3

$$\text{minimize } RSS = \sum_{i=1}^{3} \Big[y_i - \underbrace{(\hat{\beta}_0 + \hat{\beta}_1 \cdot x_1 + \hat{\beta}_2 \cdot x_2)}_{\hat{y}_i} \Big]$$

$$= \left[2 - (\hat{\beta}_0 + \hat{\beta}_1 \cdot 1 + \hat{\beta}_2 \cdot 1)\right]^2 + \left[3 - (\hat{\beta}_0 + \hat{\beta}_1 \cdot 1.5 + \hat{\beta}_2 \cdot 1.5)\right]^2 + \left[6 - (\hat{\beta}_0 + \hat{\beta}_1 \cdot 3 + \hat{\beta}_2 \cdot 3)\right]^2$$

$$\frac{\partial RSS}{\partial \hat{\beta}_0} = 6 \cdot \hat{\beta}_0 + 11 \cdot \hat{\beta}_1 + 11 \cdot \hat{\beta}_2 - 22 = 0$$

$$\frac{\partial RSS}{\partial \hat{\beta}_1} = \frac{\partial RSS}{\partial \hat{\beta}_2} = 11 \cdot \hat{\beta}_0 + 24.5 \cdot \hat{\beta}_1 + 24.5 \hat{\beta}_2 - 49 = 0$$

2 equations
3 unknowns
infinate
solutions

Consequences of multicollinearity

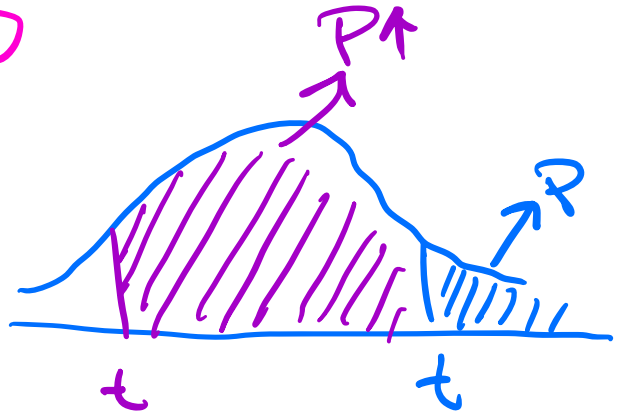
- When your predictors are perfectly correlated, there is no unique set of least square solutions.
- In real applications, it is more likely you will have predictors that are **highly correlated** (not necessarily perfectly correlated). In this case, we can still obtain unique least square solutions but there is a great deal of uncertainty in our estimates $\hat{\beta}$.
- That means the standard errors for our least square estimates could be very large.
  We can still compute least squares estimates, but the model struggles to distinguish individual effects. This results in very $\underbrace{\text{large standard errors}}_{SE(\hat{\beta})}$ for the coefficients, making them unstable.

Consequences of multicollinearity

- Reduces accuracy of $\hat{\beta}_j$ for those predictors X_j that are correlated.
- Results in increased standard errors for those $\hat{\beta}_j$'s.
- Some inference becomes problematic:

Hypothesis Test = $H_0: \beta_j = 0$, $H_A: \beta_j \neq 0$

$t = \frac{\hat{\beta}_j}{SE(\hat{\beta}_j)}$ if $SE(\hat{\beta}_j) \uparrow$ then $t \downarrow$

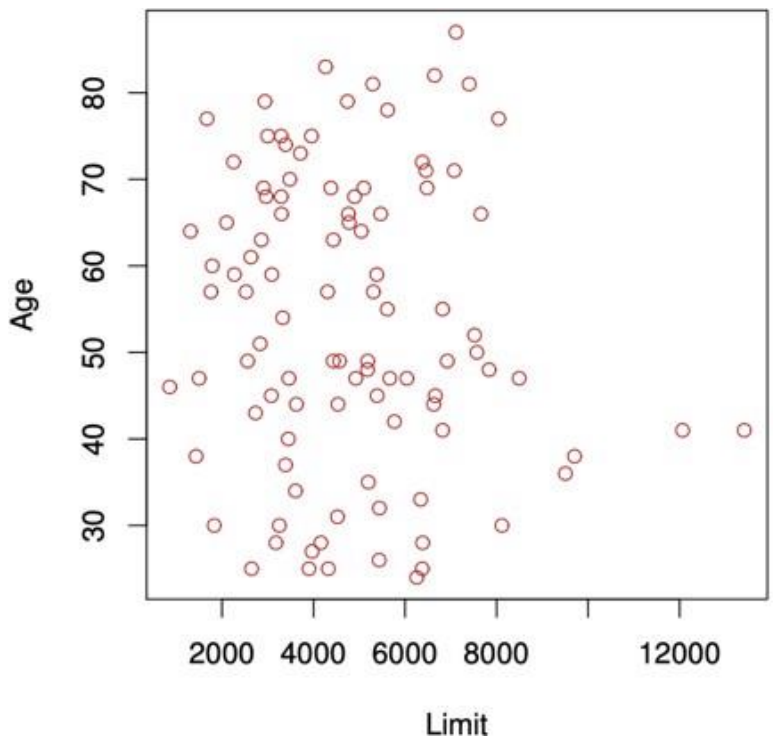


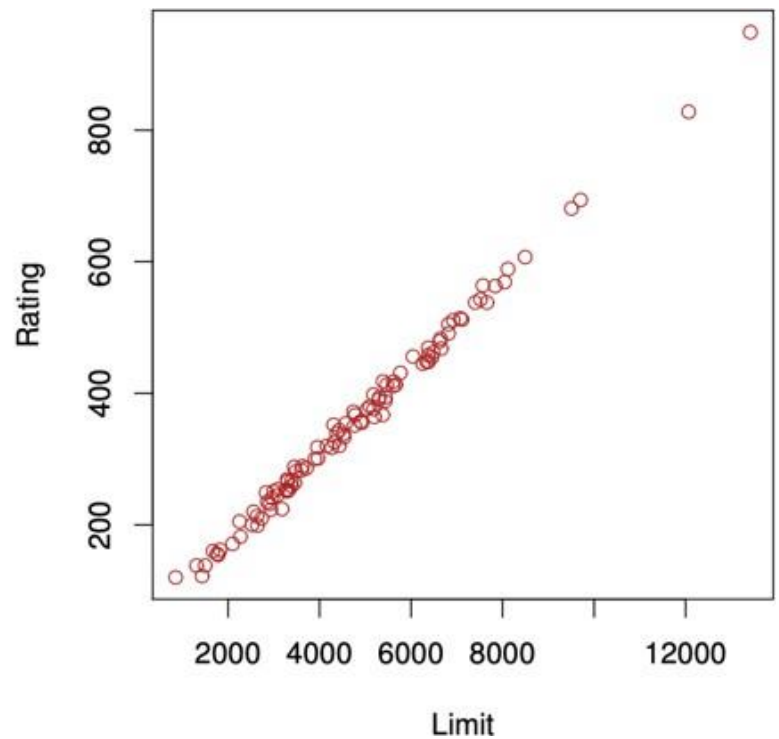
confidence interval = $\hat{\beta}_j \pm t_{\cdot} SE(\hat{\beta}_j)$

↑, less percise

prediction interval

Rating and Limit are highly correlated.





		Coefficient	Std. error	t-statistic	p-value
Model 1	Intercept	−173.411	43.828	−3.957	< 0.0001
	age	−2.292	0.672	−3.407	0.0007
	limit	0.173	0.005	34.496	< 0.0001
Model 2	Intercept	−377.537	45.254	−8.343	< 0.0001
	rating	2.202	0.952	2.312	0.0213
	limit	0.025	0.064	0.384	0.7012

How to diagnose multicollinearity?

One way is **Variance Inflation Factor (VIF)**.

- Remember that if the predictors are correlated with each other, the standard error of the regression coefficients would be larger than if the predictors were uncorrelated.
  large if correlation exists ← variance
- By this logic, we can compare the variance of β_j when fitting a full model to the variance of $\hat{\beta}_j$ if it were fit in a model by itself (SLR).

This ratio is the variance inflation factor.

VIF = larger → more correlation (in full model)

$$VIF = \begin{cases} 1 & \text{no multicollinearity (Ideal)} \\ 1 < x < 5 & \text{moderate correlation (acceptable)} \\ > 5 & \text{high multicollinearity (bad)} \\ > 10 & \text{severe multicollinearity (worse)} \end{cases}$$

R code

R code:
MLR_other_consideration_part_1.R